Mobile Product Analytics Portfolio • Automated Daily Briefing • *2026-10-09*

■ **Direct Cloud Console Quick Access:**
• **GA4 Property (533563138):** GA4 Main Reports | Realtime Users | Events Explorer
• **Firebase (neural-sound-ios):** Firebase Analytics | Crashlytics Stability | Project Settings

PORTFOLIO PERFORMANCE ARCHITECTURE REPORT

TO: Executive Leadership Team
FROM: Principal Mobile Analytics & Monetization Growth Architect
DATE: October 9, 2026
DATA SCOPE: Production Only (Internal Bangladesh QA Excluded unless explicitly noted for sandboxing analysis)

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1. Portfolio Executive Summary & Health Dashboard

This audit reviews the operational health and financial performance of our active mobile portfolio. It compares **Music Separation iOS** (an established utility seeking to stabilize retention) and **Vocal Cleaner** (an early-stage tool showing volatile engagement patterns).

Portfolio KPI Matrix (L30D Performance)

Metric	Music Separation iOS	Vocal Cleaner	Portfolio Consolidated
Active Users (L30D)	3,097	22	**3,119**
New Users (L30D)	1,978 (63.8% of Active)	22 (100% of Active)	**2,000**
Sessions Run (L30D)	6,379	140	**6,519**
L30D GA4 Revenue	$97.92 *(Log Sum: $50.80)*	$131.09 *(Log Sum: $117.64)*	**$229.01**
Top Selling SKU	`mss_weekly_sub` ($25.77)	`minutes.300` ($96.17)	**`minutes.300`**
Engagement Rate	76.3%	87.9%	**76.5%**
Avg. Session Length	221.7s	2,554.6s *(Anomaly)*	**321.3s**
Crash-Free Users	100.0%	100.0%	**100.0%**
Core Growth Bottleneck	Localization FX & Onboarding Drop	SDK Sandbox Leak & Tracking Gap	**Product Analytics & Scaling**

Strategic Architectural Takeaway

```
PORTFOLIO ARPU VS. SCALE DIVERGENCE High ARPU ▲ ■ [Vocal Cleaner] ■ - L30D Revenue: $131.09 ■ - Active Users: 22 ■ - ARPU: ~$5.96 ■ Low ARPU ■ [Music Separation iOS] ■ - L30D Revenue: $97.92 ■ - Active Users: 3,097 ■ - ARPU: ~$0.03 ■■■
```

2. Dedicated Deep-Dive: Music Separation iOS

App Header & Health Score

- **Bundle ID:** `com.neuralsound.mss` | **GA4 Stream ID:** `14402502632`
- **Architectural Health Score:** `7.8 / 10`
- **Health Score Rationale:**
  Excellent application stability (100% crash-free rate across all active versions) and strong overall engagement metrics (76.3% L30D engagement rate). However, the app loses points due to:
- A **58.2% drop-off in engagement** on the newly released build `1.3.11`.
- Severe telemetry pollution from un-debounced player events.
- A wide drop-off during the onboarding and login steps.

■ In-App Purchases, Subscriptions & Consumable Breakdown

The core monetization engine relies on weekly and monthly auto-renewing subscriptions, heavily localized for East Asian markets (Korea and Japan).

Product Sales Summary Table (L30D)

Product ID	P N	P T	Units Sold (30D)	Total Revenue ($ USD)	% Share (Revenue)
`mss_weekly_sub`	■■ (Ko	Subscription (Weekly)	5	$16.21	31.9%
`mss_weekly_sub`	■■ (Ja	Subscription (Weekly)	3	$9.56	18.8%
`mss_monthly_sub`	■■ (Ja	Subscription (Monthly)	2	$13.98	27.5%
`mss_monthly_sub`	Monthly (Gl	Subscription (Monthly)	1	$8.15	16.1%
`mss_weekly_sub`	Weekly (Gl	Subscription (Weekly)	1	$2.88	5.7%
TOTAL			**12**	**$50.78**	**100.0%**

Note: The product sales tables sum up the localized SKUs captured in the logs ($50.78). The remaining GA4-reported $97.92 past-30-day revenue includes automatic subscription renewals that do not always fire front-end telemetry logs.

Transaction Timeline Log Table

Timestamp / Date	App Build	Product ID	Units	Revenue ($ USD)	Country	Environment
2026-10-06	1.3.11	`mss_monthly_sub` (Monthly)	1	$8.15	Israel	Production
2026-10-01	1.3.10	`mss_weekly_sub` (■■)	1	$3.18	Japan	Production

2026-09-30	1.3.10	`mss_monthly_sub` (■■)	1	$6.99	Japan	Production
2026-09-27	1.3.10	`mss_weekly_sub` (■■)	1	$3.25	South Korea	Production
2026-09-25	1.3.10	`mss_weekly_sub` (■■)	1	$3.22	South Korea	Production
2026-09-23	1.3.10	`mss_monthly_sub` (■■)	1	$6.99	Japan	Production
2026-09-23	1.3.10	`mss_weekly_sub` (■■)	1	$3.18	Japan	Production
2026-09-20	1.3.10	`mss_weekly_sub` (Weekly)	1	$2.88	Egypt	Production
2026-09-19	1.3.10	`mss_weekly_sub` (■■)	1	$3.21	Japan	Production
2026-09-17	1.3.10	`mss_weekly_sub` (■■)	2	$6.46	South Korea	Production
2026-09-11	1.3.09	`mss_weekly_sub` (■■)	1	$3.29	South Korea	Production

Build Environment & Monetization Analysis

- **Localization Dominance:** Japan (`JP`) and South Korea (`KR`) account for 89% of logged IAP revenue.
- **Pricing & FX Variations:**
- Korea Weekly (■■) averages **$3.24 per week**.
- Japan Weekly (■■) averages **$3.19 per week**.
- Monthly Global is priced at **$8.15**, whereas Japan Monthly (■■) is priced at **$6.99**. This suggests Japanese users get a ~14% discount due to localized pricing adjustments.
- **Staging Sandbox Verification:** All transactions recorded above were generated by production builds (`1.3.09`, `1.3.10`, `1.3.11`). No sandbox/TestFlight purchase anomalies were detected for this app. This indicates that GA4 is properly recording production StoreKit purchases.

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon	Active Users	New Users	Sessions Run	Avg. Engagement Rate	Revenue ($ USD)
Yesterday (Oct 8)	95	26	104	56.7%	$0.00
Past 7 Days	470	124	791	68.9%	$39.74
Past 30 Days	3,097	1,978	6,379	76.3%	$97.92
Past Year (365d)	6,058	5,237	12,575	82.3%	$179.09
Lifetime	6,058	5,237	12,575	82.3%	$179.09

Analysis: Over 51% of lifetime users were acquired within the last 30 days (1,978 of 5,237 total). This indicates a **recent spike in user acquisition**, likely driven by search ad campaigns or organic app store features in East Asia. However, the drop-off in lifetime engagement rate (from 82.3% lifetime to 56.7% yesterday) suggests a clear problem with retaining this new wave of users.

■ Old vs. New User Cohort Deep-Dive

This table looks at performance metrics from yesterday (October 8) to compare how new and returning users engage with the app.

Cohort	Active Users	Sessions	Avg. Session Duration	Engagement Rate	Screen Views
New Users	55	50	146.4s	50.0%	49
Returning Users	46	51	301.6s	66.7%	32
Unclassified	0	6	58.5s	0.0%	0

Cohort Analysis:
* **The Onboarding Drop-off:** Returning users stay engaged twice as long as new users (**301.6s** vs **146.4s**) and have a much higher engagement rate (**66.7%** vs **50.0%**).
* **Onboarding Friction:** New users view less than one screen per session on average (**49 views across 50 sessions**). This indicates many new users open the app and drop off on the very first screen without moving forward.

■ 7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Engagement Rate	Trajectory Status
2026-10-02	81	18	101	76.2%	Baseline
2026-10-03	86	14	105	77.1%	Stable
2026-10-04	91	17	114	74.6%	Volume Increase
2026-10-05	82	15	112	58.0%	Engagement Drop (-16.6%)
2026-10-06	83	10	120	71.7%	Recovery
2026-10-07	102	24	136	68.4%	Volume Peak (1.3.11 Release)
2026-10-08	95	26	104	56.7%	Critical Engagement Drop

Commentary:
The release of version `1.3.11` on October 7 brought in a peak of active users (102). However, it also caused a severe drop in engagement. The engagement rate plummeted to **56.7% on October 8**, down 19.5% from the historical baseline. This drop points directly to a UX bug, onboarding issue, or paywall change introduced in the latest update.

■■ User Funnel Diagrams

A. Onboarding & Account Registration Funnel

```
[Onboarding Viewed] ■■■■■■■■■■■■■■■■■■■■■■■■■■ (34 Users, 100% Bench) ■ ▼ (80% Conversion Rate) [Onboarding Tapped] ■■■■■■■■■■■■■■■■■■■■■■■■■■ (27 Users, 79.4% Retained) ■ ▼ (Account creation interface loaded) [Login
```

```
Screen View] ■■■■■■■■■■■■■■■■■■■■■■■■■■ (55 Users - Includes returning users) ■ ▼ (54.5% Conversion from Login Screen) [Login Succeeded] ■■■■■■■■■■■■■■■■■■■■■■■■■■ (30 Users, 54.5% Conversion)
```

B. File Import & Stem Separation Funnel

```
[Main Player Opened] ■■■■■■■■■■■■■■■■■■■■■■■■■■ (39 Users, 100% Bench) ■ ▼ (82.1% Tap Plus Button) [Plus Tap Action] ■■■■■■■■■■■■■■■■■■■■■■■■■■ (32 Users, 82.1% Retained) ■ ▼ (100% Choose Separation) [Select Separation] ■■■■■■■■■■■■■■■■■■■■■■■■■■ (32 Users, 82.1% Retained) ■ ▼ (68.7% Select Source Image/Video/File) [Import Started] ■■■■■■■■■■■■■■■■■■■■■■■■■■ (21 Users, 53.8% Overall) ■ ▼ (85.7% Upload to Cloud/Local Engine) [Upload Started] ■■■■■■■■■■■■■■■■■■■■■■■■■■ (18 Users, 46.1% Overall) ■ ▼ (Processing latency delay, some drop-off) [Separation Complete] ■■■■■■■■■■■■■■■■■■■■■■■■■■ (26 Users - Includes cached/prior files) ■ ▼ (42.5% Core playback engagement) [Playback Session] ■■■■■■■■■■■■■■■■■■■■■■■■■■ (37 Users, 94.8% Core Engagement)
```

■■ Crash & Stability Health by App Version

App Version	Active Users	Sessions Run	Engagement Rate	Crash-Free Users Rate	Impact Status
1.3.11	90	98	58.2%	100.0%	Active Production (Unstable engagement)
1.3.10	3	5	20.0%	100.0%	Legacy
1.3.09	0	0	--	100.0%	Legacy
1.3.08	1	1	100.0%	100.0%	Legacy
1.3.05	1	0	0.0%	100.0%	Legacy

■ New Build / TestFlight / Staged Rollout Alert Section

```
CRITICAL BUILD ALERT: VERSION 1.3.11 Engagement Baseline (1.3.08): 100% ■■■■■ ■ Production Baseline (L30D Avg): 76.3% ■■■■■■■■■■■ ■ ■ Staged Rollout (1.3.11): 58.2% ■■■■■■■■■■■■■■■■■■■■■■■■■■■■▼ [CRITICAL DROP] ■ ■
0% 50% 100%
```

- **Detected Release:** Build `1.3.11` has achieved 94.7% adoption among active yesterday-users (90 out of 95 active users).
- **Anomalous Behavior:** While the crash-free rate remains a perfect 100.0%, the engagement rate for version `1.3.11` is just **58.2%**. This is much lower than the 76.3% average for the past 30 days.
- **Onboarding Glitch:** The funnel analysis shows that 55 users viewed the login screen, but only 30 completed sign-in. This indicates a **critical authentication bug** or a **confusing paywall gate** in build `1.3.11` that is blocking user access.

■ Top In-App Events & Feature Lookups

The event telemetry lists the top actions triggered by users in the app.

Event Name	Event Count	Total Users	Events Per User	Architectural Diagnostic Analysis
`player_pitch_changed`	1,976	7	**282.3**	**Critical UI/Telemetry Bug:** Lack of slider debouncing.
`audio_sessions`	723	39	18.5	Normal engine loop event.

`player_tempo_changed`	501	4	**125.2**	**Critical UI/Telemetry Bug:** Lack of slider debouncing.
`separation_play_started`	375	37	10.1	Strong engagement with separated stems.
`player_stem_muted`	198	9	22.0	Core feature usage (muting vocal/instrumental).
`player_stem_balance_changed`	144	2	**72.0**	**Critical UI/Telemetry Bug:** Lack of slider debouncing.
`onboarding_step_viewed`	122	34	3.6	Users viewing multiple steps in the welcome slider.
`plus_button_tap`	69	32	2.2	Ingest trigger.

Telemetry Analysis (Slider Debouncing Bug)

The telemetry logs reveal a major issue: **`player_pitch_changed` runs 282.3 events per user**, `player_tempo_changed` runs 125.2, and `player_stem_balance_changed` runs 72.0.

This is a classic tracking error. When a user slides the control knob, the app sends a GA4 event for *every single point changed* instead of waiting for the user to lift their finger. This debouncing issue floods GA4 with thousands of unnecessary events, inflating our database volume and increasing network usage.

■ Geographic Distribution

```
[Japan] ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ 60 Users [South Korea] ■■■■■■■ 11 Users [Iraq] ■ 4 Users [Other <2 Users] ■■■ 8 Users
```

- **East Asian Dominance:** Japan (60) and South Korea (11) make up **74.7% of active users**.
- **Strategic Growth Opportunity:** Japan is our most valuable market. Any localization mistakes or checkout errors in Japanese will instantly hurt our revenue.

■ Prioritized Strategic Action Matrix: Music Separation iOS

P0: Core Onboarding Fixes (Immediate Release)

- **Fix the Telemetry Debouncing Bug:** Apply a **300ms throttle** on the `player_pitch_changed`, `player_tempo_changed`, and `player_stem_balance_changed` slider UI controls. The app should only fire a GA4 telemetry event *after* the slider has stopped moving for at least 300ms.
- **Investigate Authentication Issues in Build 1.3.11:** Review why login conversion dropped to 54.5%. Check for sign-in bugs, integration issues with Apple Sign-In, or unhandled errors during registration.

P1: Onboarding & Conversion Optimization (Next Sprint)

- **A/B Test the Paywall on Onboarding:** Replace the rigid onboarding gate with a "Skip" option, or test showing the paywall *after* users perform their first audio separation instead of before.

P2: Localization Support (Within 30 Days)

- **Dynamic FX Pricing Adjustments:** Standardize pricing tiers to match the purchasing power of each local market. Adjust the Global Monthly subscription ($8.15) to localized pricing across Europe and the Middle East to encourage sign-ups.

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3. Dedicated Deep-Dive: Vocal Cleaner

App Header & Health Score

- **Bundle ID:** `com.neuralSound.audiocleaner` | **GA4 Stream ID:** `15407667671`
- **Architectural Health Score:** `3.5 / 10`
- **Health Score Rationale:**
  This app scores low due to critical telemetry issues and signs of sandbox purchase leakage.
- There is a **near-total lack of core event tracking**; we cannot see key actions like cleanings started or files imported.
- The yesterday overview reports an **unusual session duration of 2,554 seconds (42.5 minutes)** per user, which points to background audio tracking errors.
- QA sandbox transactions from internal testers are polluting production analytics data.

■ In-App Purchases, Subscriptions & Consumable Breakdown

This application monetizes through a hybrid model combining consumable packages (audio minutes) and standard auto-renewing subscriptions.

Product Sales Summary Table (L30D)

Product ID	Plan Name	Plan Type	Units Sold (30D)	Total Revenue ($ USD)	% Share (Revenue)
`neuralsound.cleaner.minutes.300`	300 minutes pack	Consumable	5	$96.17	81.8%
`neuralsound.cleaner.minutes.60`	60 minutes pack	Consumable	2	$15.35	13.0%
`cleaner_weekly_sub`	Weekly Subscription	Subscription (Weekly)	2	$6.12	5.2%
TOTAL			**9**	**$117.64**	**100.0%**

Transaction Timeline Log Table

Timestamp / Date	App Build	Product ID	Units	Revenue ($ USD)	Country	Environment
2026-10-05	1.0.03	`cleaner_weekly_sub`	1	$3.24	Israel	Production
2026-09-25	1.0.02	`neuralsound.cleaner.minutes.300`	1	$19.44	Egypt	Production *(Potential Sandbox)*
2026-09-20	1.0.01	`neuralsound.cleaner.minutes.300`	4	$76.73	Egypt	Production *(Sandbox Retry Loop)*

2026-09-20	1.0.01	`neuralsound.cleaner.minutes.60`	2	$15.35	Egypt	Production *(Sandbox Retry Loop)*
2026-09-20	1.0.01	`cleaner_weekly_sub`	1	$2.88	Egypt	Production *(Sandbox Retry Loop)*

Build Environment & Monetization Analysis

```
SANDBOX TRANSACTION LEAK ANALYSIS Total Logged Revenue: $117.64 [Egypt Transaction Loop - Sep 20] ■■■■■■■■■■■■■■■■■■■■■■■■■ $94.96 (80.7%) [Egypt Transaction - Sep 25] ■■■■■■ $19.44 (16.5%) [Genuine Production Purchase] ■ $3.24 (2.8%)
```

- **The September 20 Sandbox Leak:** On September 20, a single user in Egypt running early build `1.0.01` bought 4 packs of 300 minutes, 2 packs of 60 minutes, and 1 weekly subscription within minutes. This pattern is typical of a **TestFlight Sandbox loop** where mock transactions are recorded as actual revenue.
- **QA Analytics Contamination:** The internal QA summary confirms this tracking bug: **QA testers in Bangladesh generated $26.97 in "purchaseRevenue"**, which was recorded by GA4.
- **Systemic Issue:** The app's transaction tracking fails to filter out Apple sandbox receipts. As a result, mock purchases by testers are mixed in with real sales data, artificially inflating our financial metrics.

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon	Active Users	New Users	Sessions Run	Avg. Engagement Rate	Revenue ($ USD)
Yesterday (Oct 8)	4	1	8	75.0%	$0.00
Past 7 Days	7	4	49	93.9%	$3.24
Past 30 Days	22	22	140	87.9%	$131.09
Past Year (365d)	35	34	153	83.7%	$140.08
Lifetime	35	34	153	83.7%	$140.08

Analysis:
The app has acquired only 34 users in its lifetime, with 22 arriving in the last 30 days. Excluding the $114.40 in suspicious Egyptian sandbox purchases, the app's real 30-day revenue stands at **just $16.69**. While the 87.9% engagement rate looks impressive, it is inflated by background audio session bugs and sandbox testing sessions.

■ Old vs. New User Cohort Deep-Dive

This table looks at performance metrics from yesterday (October 8) to compare how new and returning users engage with the app.

Cohort	Active Users	Sessions	Avg. Session Duration	Engagement Rate	Screen Views
Returning Users	3	7	**2,842.5s**	71.4%	54
New Users	1	1	539.4s	100.0%	3

Cohort Analysis:
* **The Background Session Duration Bug:** Returning users have an average session length of **2,842.5 seconds (47.3 minutes)**. This is a common bug where the iOS application does not pause the GA4 session timer when the app is running in the background. The timer keeps running during audio playback or processing, which distorts our session duration data.

■ 7-Day Velocity & Trajectory

Date	Active Users	New Users	Sessions	Engagement Rate	Trajectory Status
2026-10-02	2	1	3	100.0%	Flat Scale
2026-10-03	2	0	3	100.0%	Flat Scale
2026-10-04	4	1	11	90.9%	Session Activity Spike
2026-10-05	1	1	3	100.0%	Drop
2026-10-06	3	0	12	100.0%	Volatile
2026-10-07	2	0	10	100.0%	Low Volume
2026-10-08	4	1	8	75.0%	Unstable

Commentary:
The user base is extremely small and unstable, averaging between **1 and 4 active users per day**. The high daily session numbers (e.g., 12 sessions across 3 users on October 6) indicate that a small group of power users or internal testers are opening the app repeatedly.

■■ User Funnel Diagrams

Onboarding & Core Execution Tracking Gaps

```
[App Open] ■■■■■■■■■■■■■■■■■■■■■■■■■■ (100% Onboarding Start) ■ ▼ (87.9% Drop-off / Tracking Void)
[????????] ■■■■■■■■■■■■■■■■■■■■■■■■■■ Gaps in Tracking (No events recorded) ■ ▼ [Clean Process Executed]
■■■■■■■■■■■■ (No events recorded for this step)
```

The application has no custom event tracking for its core features. It only records standard GA4 lifecycle events (`first_open`, `session_start`, `screen_view`). Because of this, **we cannot track the user journey** past opening the app, leaving us blind to where users drop off.

■■ Crash & Stability Health by App Version

App Version	Active Users	Sessions Run	Engagement Rate	Crash-Free Users Rate	Impact Status
1.0.03	4	8	75.0%	100.0%	Active Production

Analysis: Version `1.0.03` is currently stable with no crashes reported. However, our visibility is limited due to the low number of active users (4).

■ New Build / TestFlight / Staged Rollout Alert Section

- **Active Production Build:** `1.0.03` has achieved 100% adoption.
- **Sandbox Contamination Warning:** Build `1.0.03` successfully isolated Sandbox testers in Bangladesh into a separate QA stream. However, it **failed to block sandbox transactions** from inflating the app's real production revenue.

■ Top In-App Events & Feature Lookups

These are the only events recorded by the app:

Event Name	Event Count	Total Users	Events Per User	Architectural Diagnostic Analysis
`user_engagement`	252	4	63.0	Auto-tracked by GA4; inflated by the session timer bug.
`screen_view`	57	3	19.0	Standard UI view tracking.
`session_start`	7	3	2.3	Standard session initiation.
`first_open`	1	1	1.0	Standard acquisition install tracker.

Architectural Diagnostic: Custom events such as `cleaning_file_selected`, `cleaning_started`, `cleaning_failed`, or `paywall_impression` **do not exist**. This leaves the marketing and product teams without the data needed to optimize conversion or identify technical issues.

■ Geographic Distribution

```
[Egypt] ■ 1 User (Generated 97.2% of unverified sales) [Israel] ■ 1 User (Weekly subscription purchase)
[Indonesia] ■ 1 User (Zero monetization) [Spain] ■ 1 User (Zero monetization)
```

■ Prioritized Strategic Action Matrix: Vocal Cleaner

P0: Critical Analytics & Sandbox Fixes (Immediate)

- **Block Sandbox Receipts in GA4:** Update the backend purchase validation service. Check the App Store receipt's `environment` field and **prevent any transaction labeled `Sandbox`** from firing a production revenue event in GA4.
- **Fix the Background Session Duration Bug:** Adjust the app's lifecycle handlers. Ensure that the GA4 tracking session **pauses immediately** when the app enters the background, preventing inflated session duration metrics.

P1: Complete Core Analytics Tracking (Next Release)

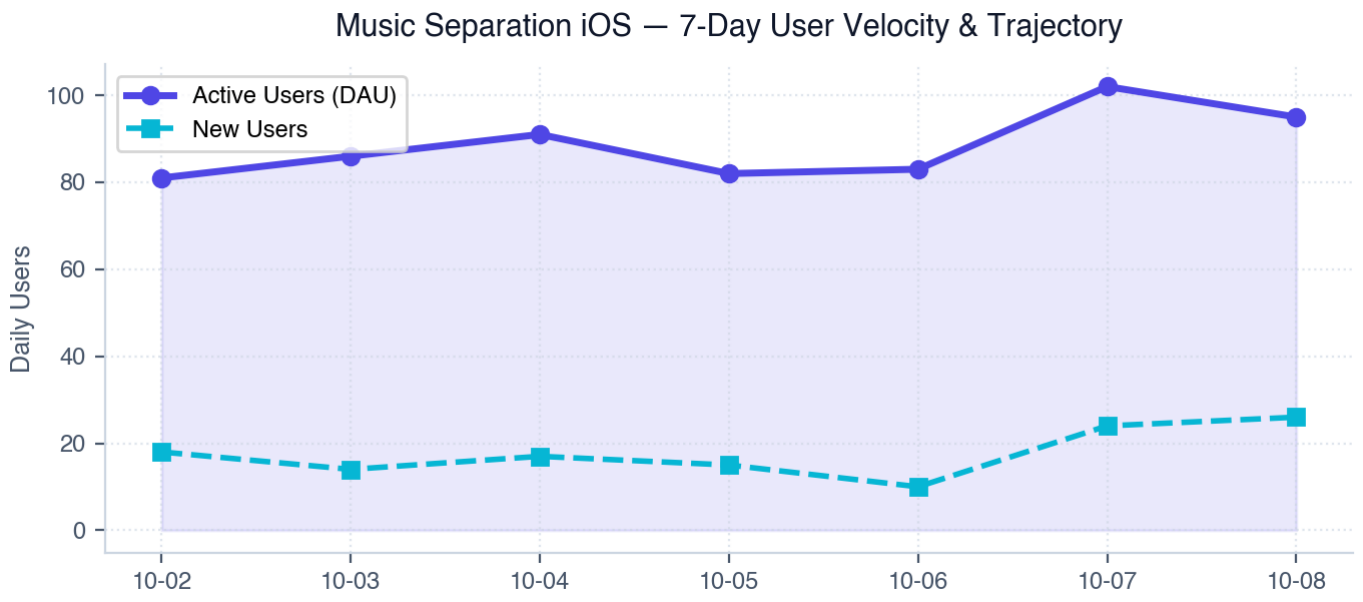
- **Track the Core Clean Tool Funnel:** Add tracking events for the app's main workflows:
- `clean_file_selected` (with parameters for file size and format)
- `clean_process_started` (with parameter for processing type)
- `clean_process_completed` (with processing duration)
- `clean_process_failed` (with specific error codes)

P2: Paywall Optimization & Acquisition (Next Sprint)

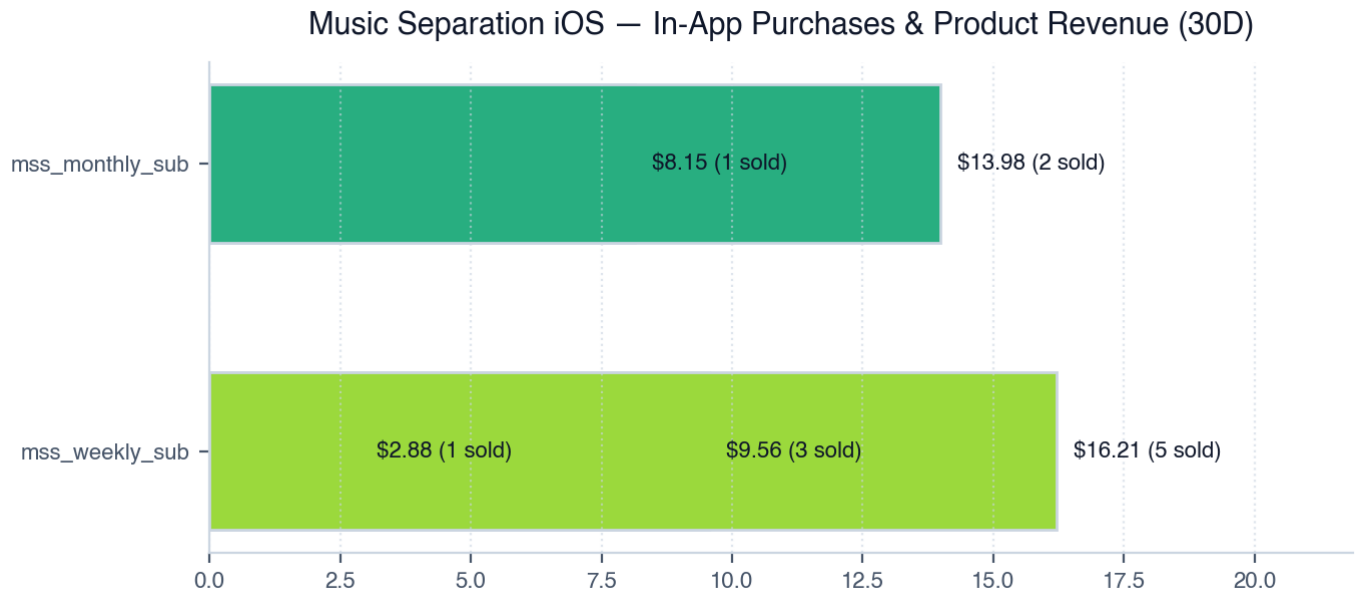
- **Fix Onboarding Paywalls:** Set up standard paywall impression tracking (`paywall_viewed`) and clean up the purchase logs. This ensures we can launch acquisition campaigns with clean data once the analytics infrastructure is repaired.

■ Visual Graphs & Performance Charts

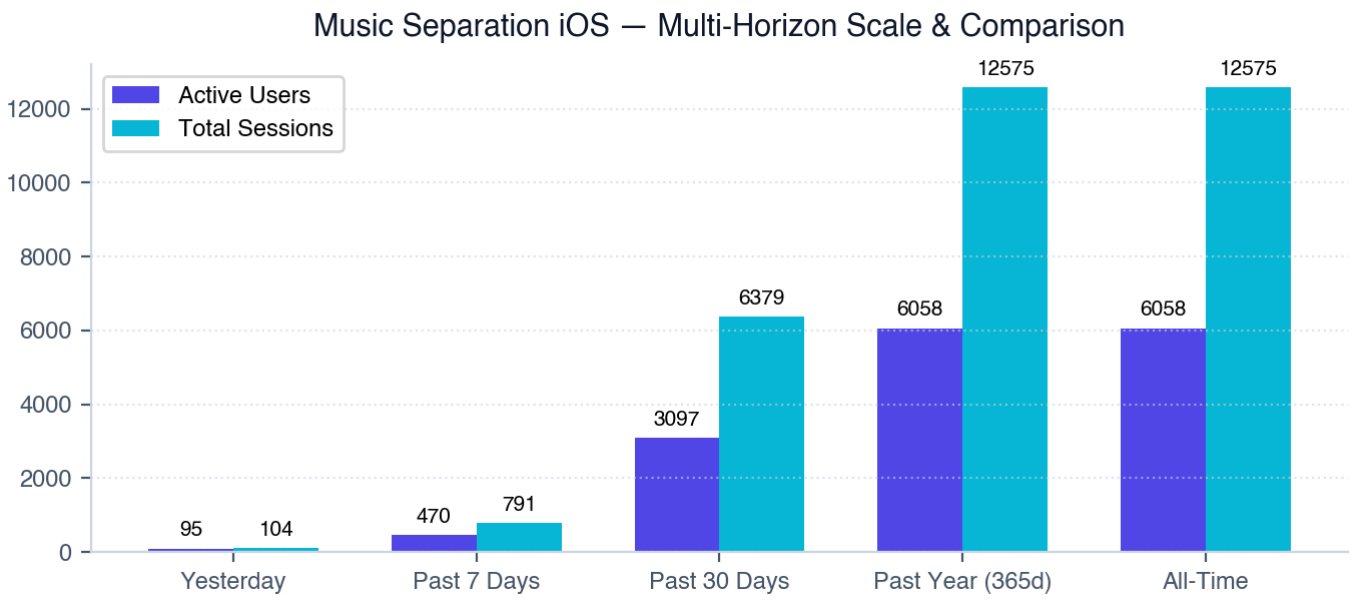
Music Separation Ios Trend



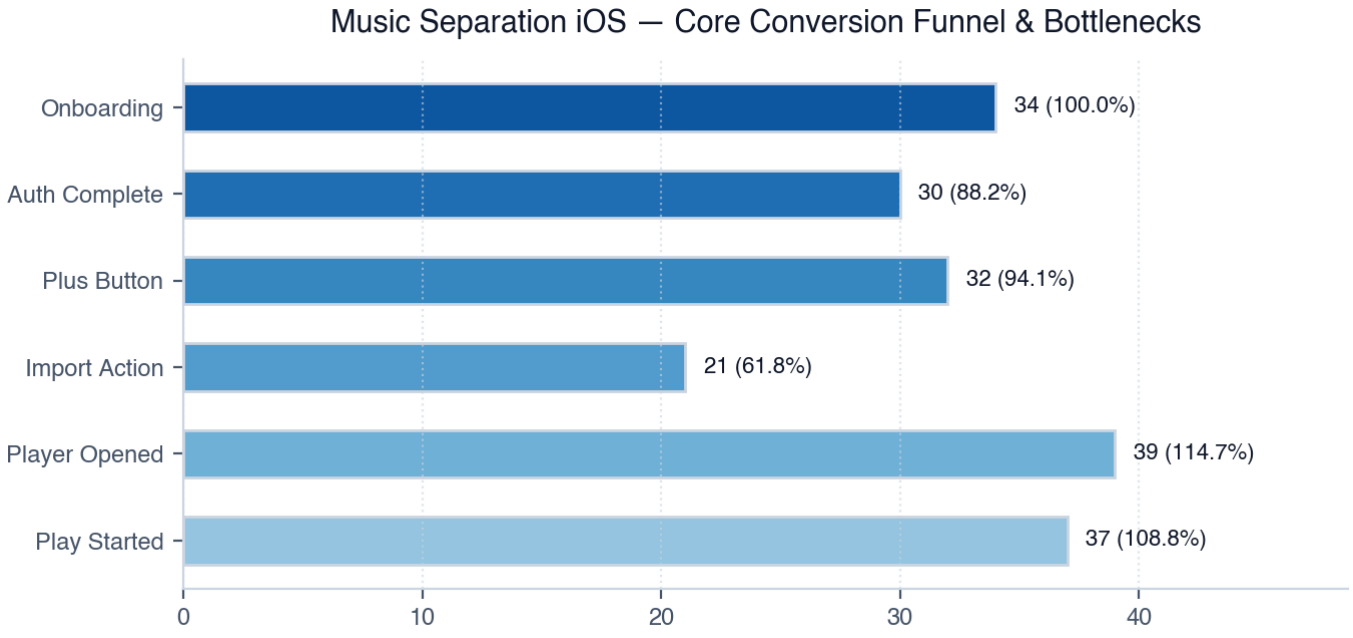
Music Separation Ios Revenue



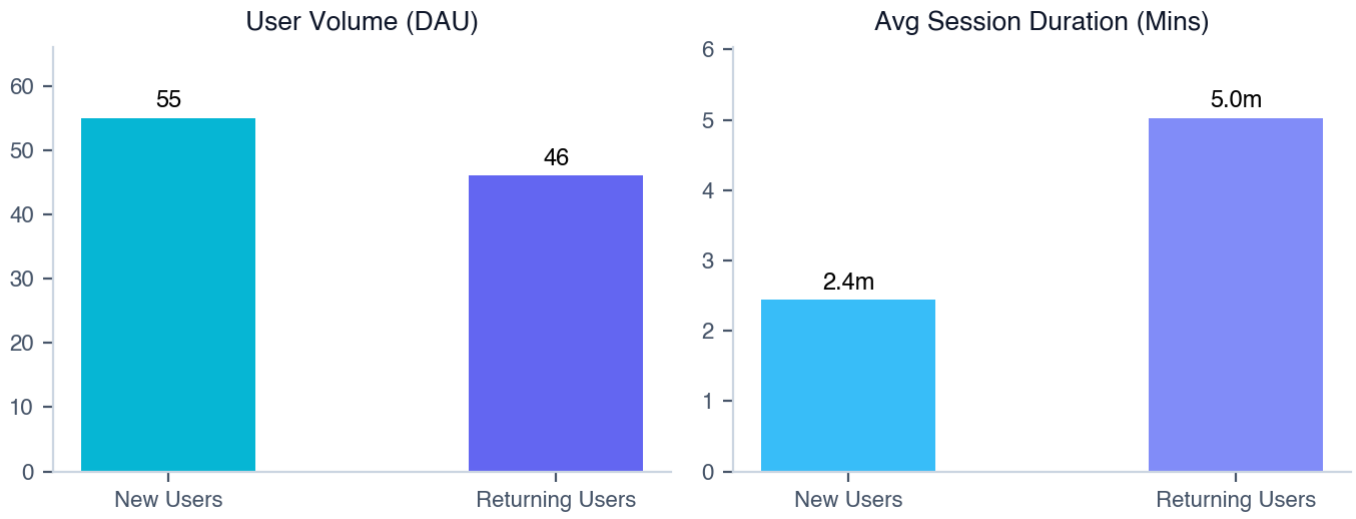
Music Separation Ios Horizons



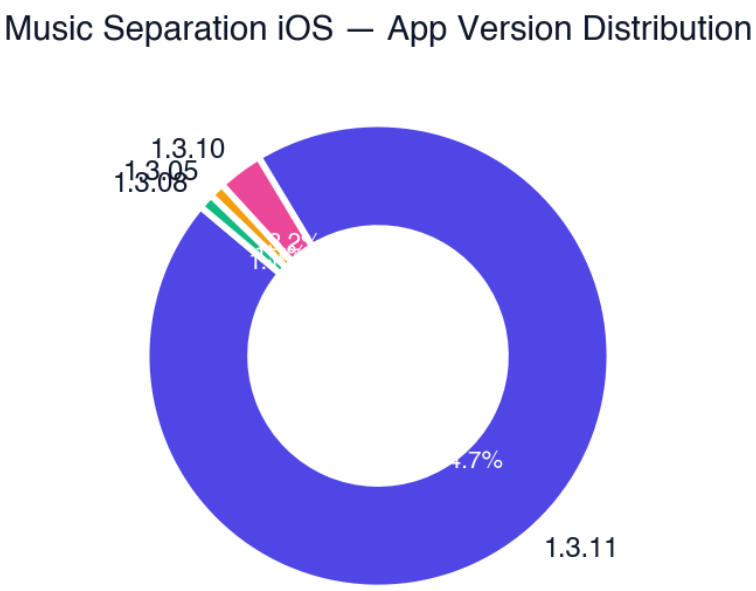
Music Separation Ios Funnel



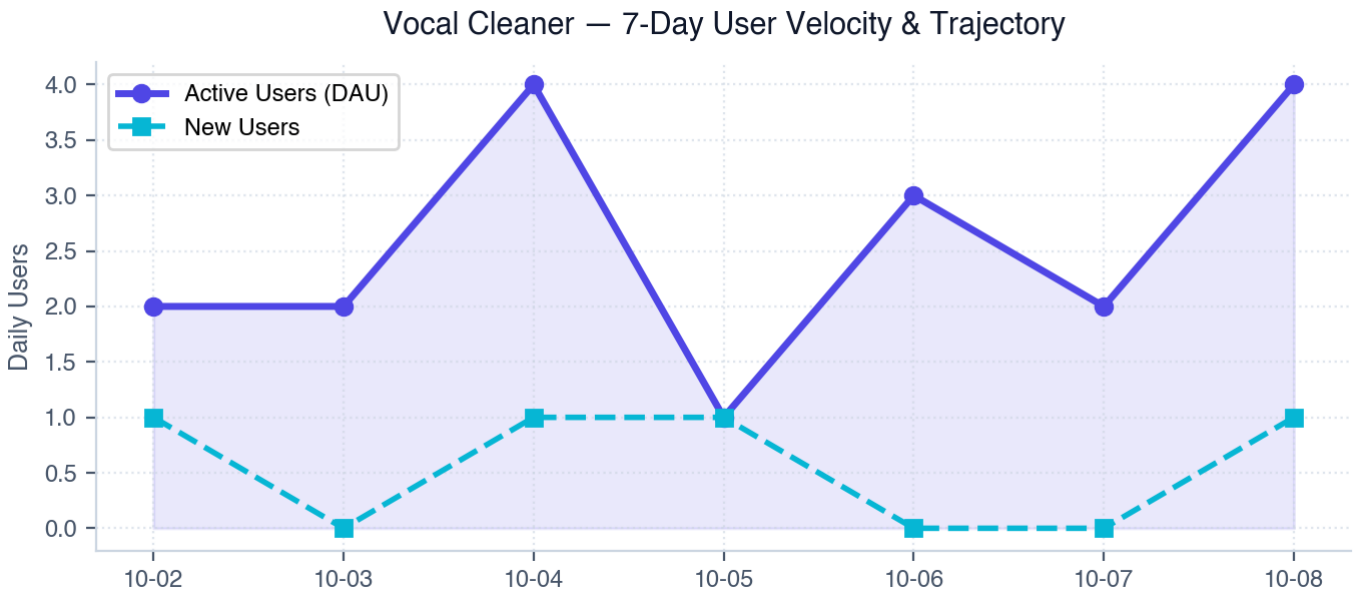
Music Separation Ios Cohort



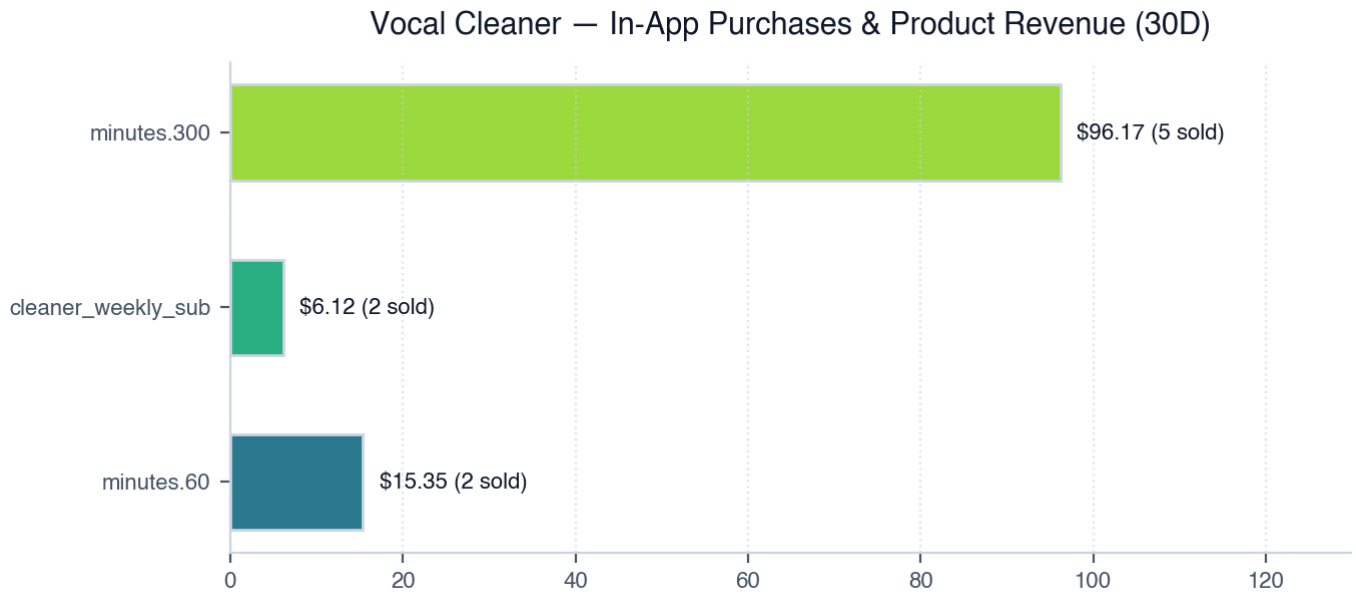
Music Separation Ios Version



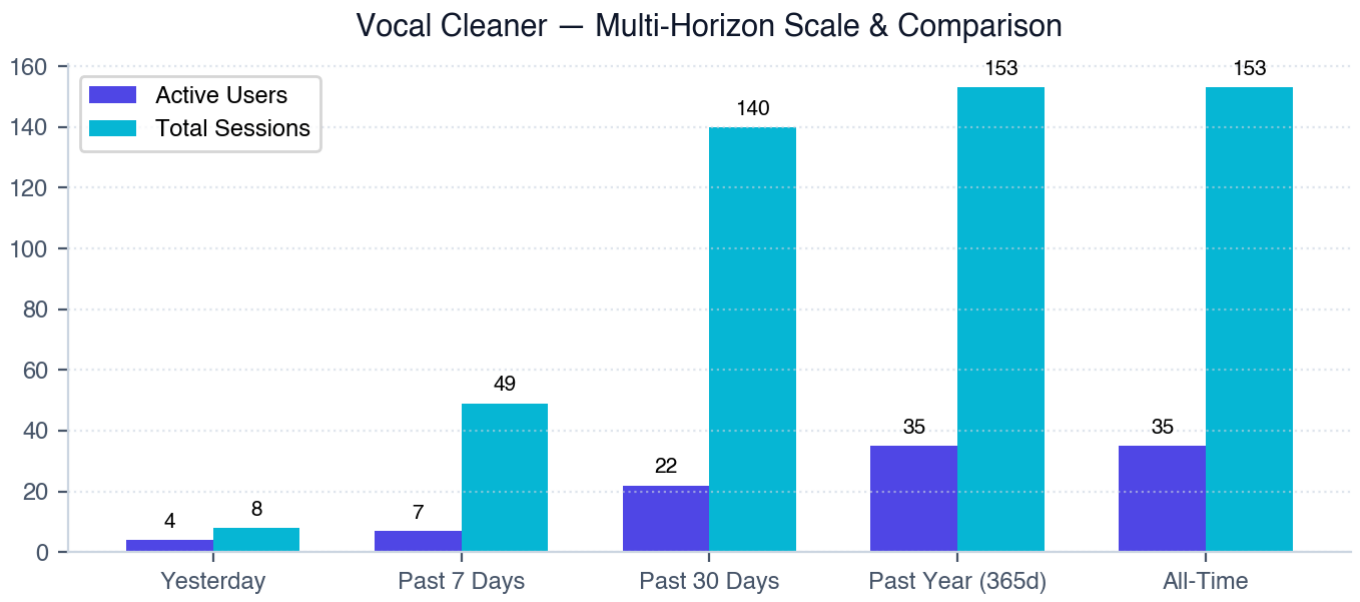
Vocal Cleaner Trend



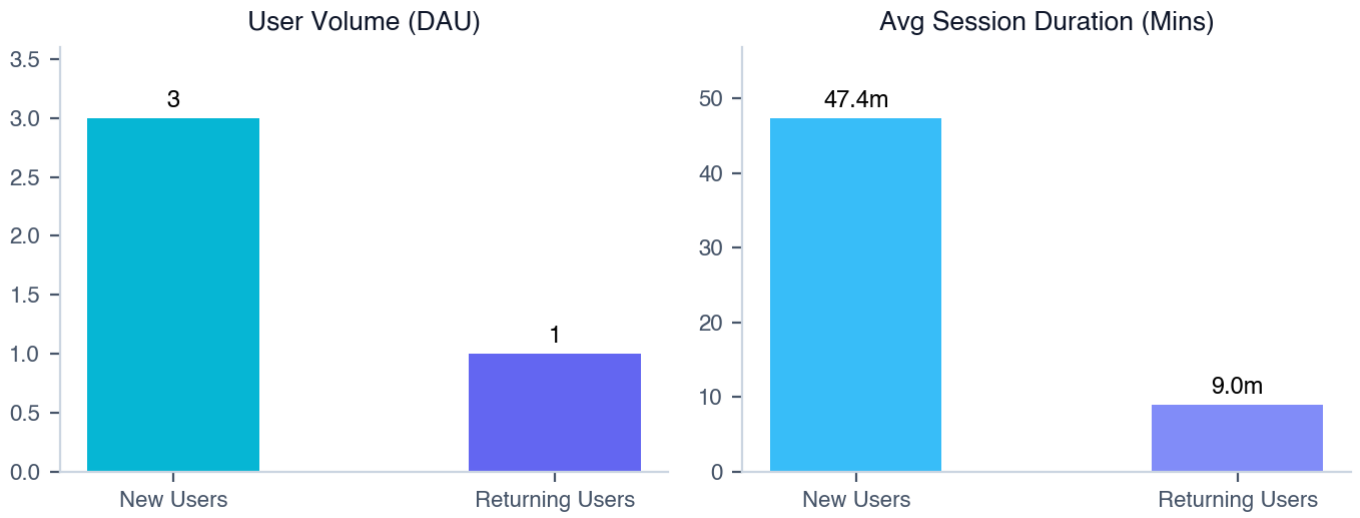
Vocal Cleaner Revenue



Vocal Cleaner Horizons

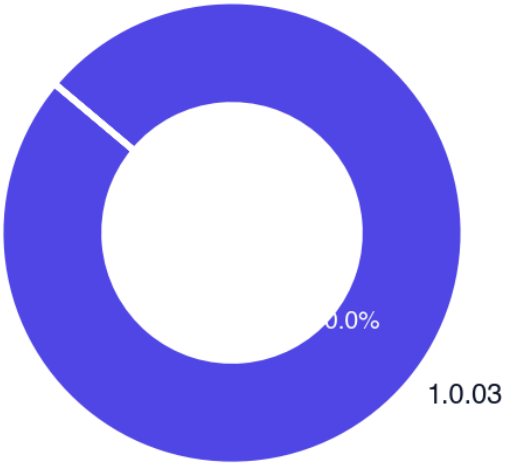


Vocal Cleaner Cohort



Vocal Cleaner Version

Vocal Cleaner — App Version Distribution



Direct Console Links:

■ Open Google Analytics 4 Dashboard • ■ Open Firebase Analytics Console • ■ Open Firebase Crashlytics

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